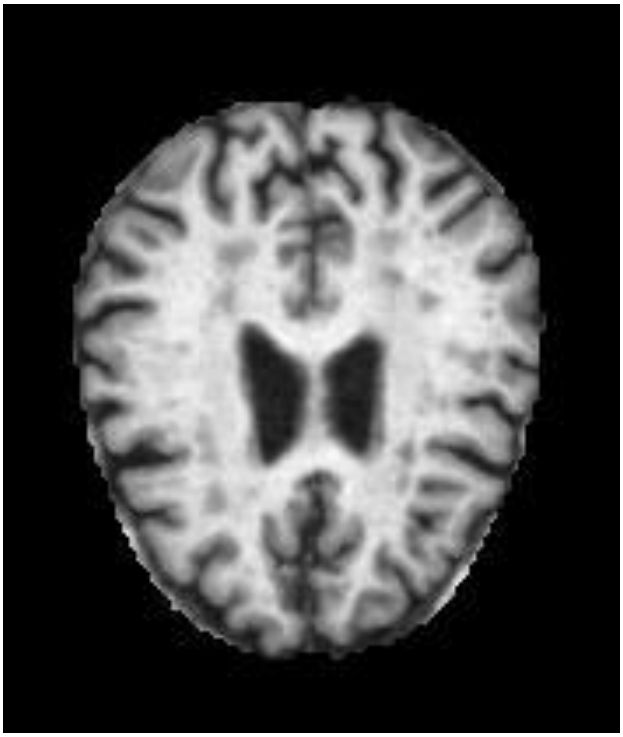

Figure 1- MRI image of a brain found in the Kaggle Dataset used [1]

Haider Ahmed | CO3201 Computer Science Project

Alzheimer's Disease identification based on convolutional neural networks.

Dissertation

“School of Computing and Mathematical Sciences, University of Leicester”

DECLARATION All sentences or passages quoted in this report, or computer code of any form whatsoever used and/or submitted at any stages, which are taken from other people’s work have been specifically acknowledged by clear citation of the source, specifying author, work, date, and page(s). Any part of my own written work, or software coding, that is substantially based upon other people’s work, is duly accompanied by a clear citation of the source, specifying author, work, date, and page(s). I understand that failure to do this amounts to plagiarism and will be considered grounds for failure in this module and the degree examination as a whole.

Name: Haider Ahmed

Signed: HA

Date : May 2024

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Chapter 1 - Abstract:

This project develops and evaluates a Convolutional Neural Network (CNN) model designed specifically for identifying and diagnosing Alzheimer's disease using magnetic resonance imaging scans (MRI). The recent developments in machine learning, specifically in the application of CNNs, present a novel approach to medical diagnosis, as it offers potential for significant improvements in accuracy and efficiency over traditional methods.

This project was driven by the need to enhance diagnostic procedures for Alzheimer's disease which affects millions globally and poses challenges in early and accurate detection. This project utilises a widely recognised dataset from Kaggle, which includes diverse stages of Alzheimer's represented in MRI scans, the project involved comprehensive data preprocessing to ensure the quality and uniformity of the data for efficient model training.

The custom CNN architecture developed in this project was tailored to effectively extract pivotal diagnostic features from the MRI scans. The model demonstrated high accuracy, precision and recall through rigorous training, testing and validation phases. which indicated its ability to distinguish between different stages of Alzheimer's disease effectively

Despite the challenges of overfitting and class imbalance, strategic implementations of regularization techniques, class weights and advanced data augmentation were employed to refine the model's performance, additionally, adaptive learning rate adjustments were used to enhance the model's generalization capabilities.

This conclusion draws from this project and the potential of CNNs in revolutionising Alzheimer's diagnostics. The model not only fulfils the technical expectations by achieving high diagnostic performance but also sets the stage for future research and application in clinical settings. These findings also advocate for further integration of machine learning into healthcare with a view towards developing more sophisticated models that can be deployed in the real world.

Chapter 2 - Introduction:

Context and Background:

Alzheimer's disease (also known as AD) is a progressive neurodegenerative disorder that predominantly affects the elderly, and can lead to severe cognitive impairment and memory loss. As populations age worldwide, the prevalence of Alzheimer's continues to rise, presenting a significant challenge to healthcare systems across the globe. Early and accurate diagnosis of Alzheimer's can be a critical method to effectively manage and treat the disease, potentially delaying the progression of symptoms and improving the quality of life for patients.

Advancements in technology, specifically in the fields of artificial intelligence (also referred to as AI) and machine learning, have introduced new possibilities for enhancing diagnostic methods for AD. Among these technologies, Convolutional Neural Networks have shown exceptional promise in interpreting medical imaging to detect and diagnose a variety of diseases, including Alzheimer's. These models can analyze complex imaging data with a level and speed that can surpass the traditional methods used for diagnosis.

Project Aims and Objective:

The primary aim of this project is to develop and implement a CNN model that can accurately identify early Alzheimer's in patients using magnetic resonance imaging (MRI) scans. The objectives set to achieve this include:

1. Data gathering and Preprocessing: to collect a comprehensive dataset of MRI images of the brain relevant to Alzheimer's disease and perform necessary preprocessing methods to prepare the data for efficient and effective CNN modelling. This will involve procedures such as data augmentation, normalizing data and standardizing image formats.
2. Model Development and Design: To design and build a CNN that is specifically tailored for the task of image-based Alzheimer's disease diagnosis. This may involve selecting an appropriate architecture that can process and learn from the complex features within the MRI scans.
3. Model Training and evaluation: To train the CNN using the prepared dataset and rigorously evaluate its performance using a variety of metrics, such as accuracy, precision, recall, and area under ROC. This step also includes fine-tuning the model to enhance its diagnostic capabilities.
4. Addressing the Challenges in AI modelling: To tackle common challenges in machine learning such as overfitting, class imbalance and model interpretability. This will involve but is not limited to strategies like regularization, class weight, adjustment and possibly exploring more complex model architectures.

By achieving these objectives, the project will not only contribute to the technical field but will also address the critical social need for enhancing the diagnostic capabilities for Alzheimer's disease. In summary, this project seeks to harness the power of advanced machine learning techniques to improve the diagnosis of Alzheimer's disease, demonstrating the potential to transform healthcare practices using AI.

Chapter 3 – Survey of Literature

3.1 - Alzheimer's disease.

In recent times due to the advancement of Machine Learning methods, there has been a significant shift in the way Alzheimer's Disease (Also known as AD) is identified and diagnosed, this is largely due to the application of techniques such as Convolutional Neural Networks (CNNs). Memory loss and cognitive decline are hallmarks of Alzheimer's disease, the most common form of dementia. These symptoms are brought on by the gradual death of brain cells, which is the cause of Alzheimer's disease. These changes in brain structure can be observed in great detail using advanced imaging techniques such as structural magnetic resonance imaging (sMRI), functional magnetic resonance imaging (fMRI), positron emission tomography (PET), and diffusion tensor imaging (DTI). These techniques provide essential information for diagnosis. 2 [2][3][4]:

3.2 – Imaging Technologies

The development of medical imaging has significantly improved the ability to detect and diagnose Alzheimer's disease. This is because imaging technologies and data acquisition have advanced. To detect neurological changes that are associated with Alzheimer's disease (AD), techniques such as magnetic resonance imaging (MRI) and positron emission tomography (PET) scans are essential. To train CNNs to diagnose and predict Alzheimer's disease with a high degree of accuracy, resources such as the Alzheimer's Disease Neuroimaging Initiative (ADNI) and public datasets from platforms such as Kaggle are extremely valuable. These resources provide a large amount of diverse data, which can be essential for the training of CNNs. (5) [5](6) [6]

Data preprocessing is an essential step in the process of preparing imaging data for CNN analysis. This step includes converting the format of the data and performing data preprocessing. It is possible, for example, to convert images from the ADNI databases into more common formats such as JPEG to improve their compatibility with various CNN models. To make data usable for algorithms and ensure consistency across a variety of datasets, this standardization is essential. [7]:[8]:

3.4 – Rapid Advancements in Feature Extraction Methods

The field of Alzheimer's disease research has witnessed substantial progress in the area of feature extraction techniques. Traditional methods such as voxel-based morphometry and tensor-based morphometry are gradually being supplanted by convolutional neural networks (CNNs). CNNs can automate and improve the extraction of complex features from brain images, which ultimately leads to an improvement in diagnostic accuracy. [3]:[8]:

The Techniques for Integrating Data, Section 3.5:

The accuracy of Alzheimer's disease classification can be significantly improved by integrating data from a variety of imaging sources. The diagnostic view of the impact that Alzheimer's disease has on the brain has been improved thanks to the utilization of techniques such as canonical correlation analysis (CCA), which has been used in recent times to combine data from various imaging types such as sMRI and PET.[9] [9]

3.6 - Convolutional Network Architectures

The architecture of a CNN is extremely important for the successful classification of AD. To maximize the amount of information that can be learned from data, researchers make it possible for them to experiment with different configurations of these networks by adjusting the layers, filter sizes, and depths. In addition to assisting the models in effectively learning from training data, these adjustments also assist the models in generalizing well to data that they have not previously received. [10] [10][11](17) [17]

3.7 - Overfitting

Addressing the issue of overfitting is a common challenge that arises during the training of CNNs, particularly when larger datasets are utilized. When a model learns the training data too well, including the nose and errors, it can degrade its performance on newly introduced data. This happens when the model learns the training data too well. To improve the model's capacity to generalize, various methods such as cross-validation and data augmentation are utilized.

3.8 – Utilising Three-Dimensional Convolutional Neural Networks

The utilization of three-dimensional CNNs in Alzheimer's disease detection is especially promising due to the three-dimensional nature of brain data. The ability of these models to capture spatial hierarchies in brain images is improved as a result of the use of layers 3.9-Recent Developments

The most recent advancements in the field of medical imaging analysis include hybrid models that combine convolutional neural networks (CNNs) and transformer networks. These hybrids solve certain problems that arise when analyzing complex regions within images, which will, in turn, lead to an improvement in the detection and segmentation of intricate patterns in comparison to conventional models.

Chapter 4 - Design and Implementation :

4.1 – Data Collection and Preparation

Data Sourcing:

This project makes use of a structured dataset from Kaggle that was developed expressly for the categorization of Alzheimer's disease. This dataset has many classes of demented brains, ranging from brains that are not affected by Alzheimer's disease to brains that are very mildly affected by the disease, and it is presented in the form of MRI scans. In addition, the dataset has already been partitioned into training and test datasets, which makes it possible to train and validate models easily.

This dataset was initially available in a pre-partitioned format, with distinct sets for training and testing, this greatly boosts the speed at which training and validation occurs, as there is no need to separate data. Alongside this, the supersaturation ensures that the data can be trained on a substantial amount of data and subsequently tested on unseen data, which is crucial for verifying the model’s ability to generalize beyond the training samples

My decision to use this specific dataset over others was heavily influenced due to its popularity as within the machine learning community, it has been extensively used by other researchers and practitioners. Those of which have applied a variety of machine learning models to it. Reviewing these exciting works provided insights into the data reliability and potential challenges and successes I might encounter. This gave me a solid foundation for anticipating the behaviour of different models and understanding the nuances of the dataset better.

Furthermore, the widespread usage of these datasets has also resulted in a wealth of shared knowledge and community feedback. Many users have documented their methodologies, results and even pitfalls, each of which offers a significant learning experience for someone starting like myse;f. By analyzing these prior projects, I was able to identify common issues like overfitting and class imbalances and how they tackled these issues, which informed the strategies I employed in my model development.

The choice was also strategic from a novelty perspective. While many available models had their versions of the CNN Architectures, my project aimed to innovate and differ from these models. The familiarity of other researchers with this dataset allowed me to position my work within a known context, while also making my contributions and deviations more impactful and easier to understand and evaluate. I aimed to build upon the collective knowledge while also introducing unique elements through advanced image processing techniques and neural network modifications specifically for Alzheimer's disease.

By choosing a dataset that is both well-understood and commonly used, I not only leveraged existing knowledge but also contributed to an ongoing practical dialogue in Alzheimer’s disease research through machine learning.

Data Preprocessing:

To prepare the images for use in neural networks, we must first go through the process of preparing them for CNN analysis. It has been achieved by.

Resizing: To preserve uniformity in the size of the input data, the images have been resized to a size of 224 by 224 pixels.

Normalization: to promote more efficient training, the pixel values are modified to ensure they fall within the range of [0,1].

Augmentation: techniques such as rotations and flips are used with the goal expand the training dataset and assist the model in better generalizing to the more recent data. This is known as "augmentation."

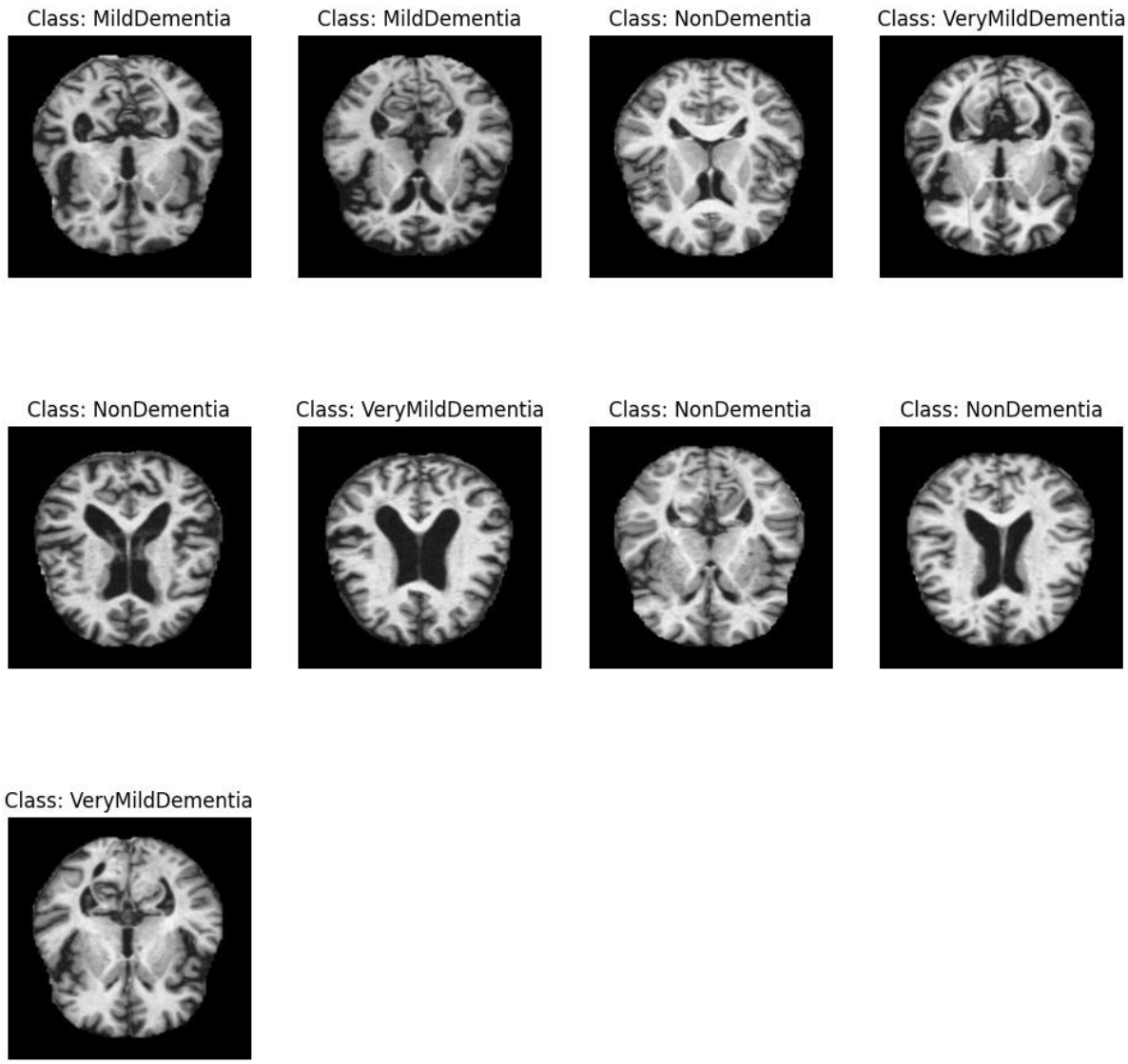

Figure 2 - An example of each class of image post processed

4.2 – Model architecture and optimization:

Custom CNN –

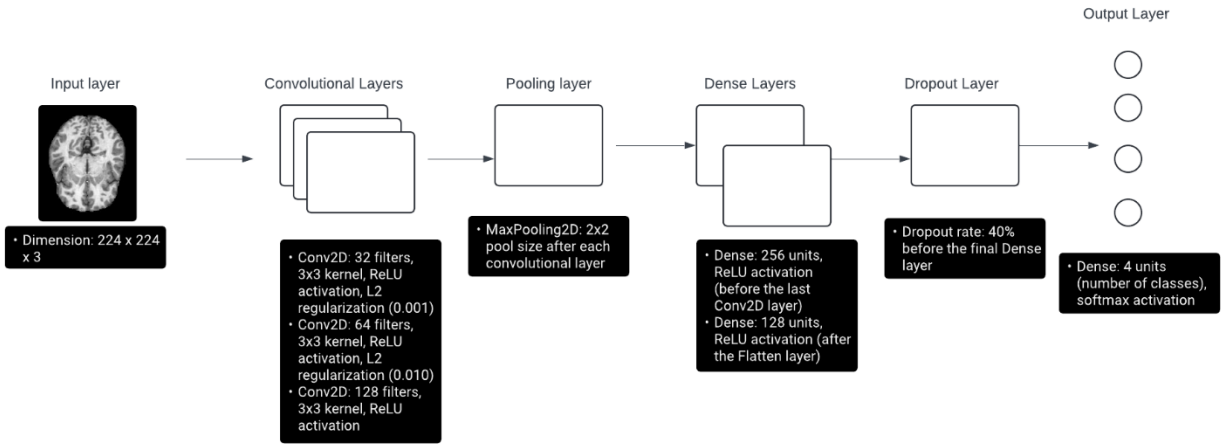

Figure 3 - A diagram showing the structure of the custom CNN

I have developed a specialized architecture to properly process brain scan pictures to construct this CNN, which is a custom CNN that I constructed just for this project. The CNN is designed to successfully evaluate brain scan images to identify the various phases of Alzheimer's disease. I accumulated my information from a collection of sources that were found across the internet. This is accomplished through the utilization of the TensorFlow Keras API, which makes it possible to develop a model layer by layer. Layer Configuration:

Convolutional Layers: The network of convolutional layers is composed of numerous layers, which are referred to as "Conv2D." Each of these layers involves the utilization of filters to capture spatial hierarchies of data, ranging from simple edges to more complicated patterns. In addition, the network can acquire the ability to learn complex transformations by the utilization of ReLu (Rectified Linear Unit) activation, which guarantees non-linear relationships.

Pooling Layers: After each convolutional layer, there is a 'MaxPooling2D' layer that comes after it. This layer minimizes the spatial dimensions of the output, which in turn reduces the computational complexity and the number of parameters.

Regularization Techniques: The network comes to a close with a completely connected set of layers, which are referred to as "Dense" layers. These levels incorporate the features that were learned from the layers that came before them to create final predictions. This layer is essential since it combines the features that have been learned into predictions for each class

Output Layer :

The final output layer uses a softmax activation function to provide a probability distribution across four diagnostic categories, translating the network’s computations into actionable predictions.

Training and Optimization:

Compiling the Models:

The model is compiled with the Adam optimizer, this optimizer is known for its efficiency in handling sparse gradients on noisy problems. The learning rate is set in consideration of convergence speed and training stability.

The loss function used is categorical cross-entropy, which is ideal for multi-class classification problems where each class is mutually exclusive.

Training Approach:

The model is trained using image data generators which facilitate real-time data augmentation and preprocessing, which is crucial for enhancing model generalization. Training involves multiple epochs where the model learns from incrementally adjusted weights to minimize loss on a batch-by-batch basis.

The ReduceLROnPlateu was also used to dynamically reduce the learning rate when improvements in validation loss stall, helping fine-tune the model toward optimal solution and prevent overfitting.

Evaluation Metrics:

Performance is monitored using a suite of metrics including accuracy, precision, recall, and AUC. These Metrics provide a comprehensive view of the model's effectiveness across the different stages of Alzheimer's.

4.3 - Web Application Development :

Streamlit Web Application:

User-Friendly Interface – Developed using Streamlit, this application provides an interactive platform for its users to upload brain scans and receive predictions on Alzheimer's disease stages. I have built this interface to be intuitive, and to cater to users with varying levels of technical expertise.

Functional Layout – The application features multiple pages, including Predictions, Model Performance, and FAQ, these pages are designed to educate and cater to different aspects of user interaction and educational needs.

Prediction Page:- This page allows users to upload brain scan images and see the model's predictions for different stages of Alzheimer's disease. Users can also adjust the confidence threshold, which lets them control how certain the model should be before making a prediction. This feature enhances the user's ability to tailor the app's responses to their needs.

Model Performance Page -Here, the application demonstrates how the model predicts Alzheimer's disease stages by showing simulated outcomes. This page helps users grasp how the model processes images and assigns probabilities to each disease stage, making the model's decision-making process more transparent and understandable.

FAQ Page - This section provides comprehensive information about the different stages of Alzheimer's disease. It aims to educate users by offering clear, detailed descriptions of each stage, thereby enriching their understanding of the disease and its progression.

Chapter 5 - Results and Discussion:

5.1 Detailed Analysis of Training Process:

Model Architecture:

Custom CNN: Designed specifically for Alzheimer's disease classification, this model included several layers of convolutional networks intended to capture features at various levels of abstraction from brain scans. The architecture was relatively shallow compared to ResNet50, aimed at reducing computational demand while still capturing essential features.

Data Preparation:

The Model was trained on a dataset of brain scans which were pre-processed to standardize image size and pixel values. This step is important as it ensures that the model learns from features relevant to the disease rather than artefacts introduced by differing image scales or intensities.

Augmentation Techniques: To enhance the robustness of the models and to prevent overfitting, data augmentation strategies such as rotation, zoom, and horizontal flipping were used. This not only artificially expands the training dataset but also helps the model generalize better by learning from slightly altered variations of the same images.

Batch Size and Learning Rates:

Determining the optimal batch size was crucial for effective learning. Larger batches provide a more accurate estimate of the gradient, but they require more memory and can lead to a less generalizable model. Furthermore, smaller batches, while more noisy in estimating the gradient, allow for more frequent updates and potentially faster learning.

The learning rate was initially set high to allow for rapid convergence and gradually reduced using the ReduceLROnPlateau strategy. This approach adjusts the learning rate when the improvement plateaus, helping to fine-tune the model parameters and avoid overshooting the minima.

Hyperparameter Selection:

The selection of hyperparameters such as the number of layers, the size of the filters in convolutional layers, and dropout rates was based on preliminary experiments that gauged their impact on the model's performance on a validation set.

Regularization Techniques: To mitigate the risk of overfitting, both models employed techniques like dropout and L2 regularization. These methods are particularly important in medical imaging, where the model must perform well across unseen data that may significantly differ from the training set.

Training Strategy:

Epochs and Validation: The models were trained over multiple epochs, with performance on a held-out validation set monitored after each epoch. This ongoing validation serves as a check against overfitting and provides insight into when the models begin to plateau in terms of learning new features.

Callbacks: Besides learning rate adjustments, other callbacks were utilized to monitor training, including early stopping which terminates training if the validation performance degrades for a set number of consecutive epochs.

Handling Imbalanced Data:

One of the significant challenges was the imbalance in the dataset, with some stages of Alzheimer's being underrepresented. This was addressed through strategic data augmentation for underrepresented classes and the use of class weighting in loss calculations to give more importance to rare classes during training.

Computational Constraints:

Given the extensive computational demands, especially of the ResNet50 model, training was optimized by employing mixed-precision computations, which reduce memory

Model Evaluation and Refinement:

Throughout the training process, the models were continually evaluated against the validation set to gauge their performance. Insights gained from these evaluations were used to iteratively refine the model architectures and training strategies.

Feedback Loop: This iterative process was crucial in developing models that not only perform well on the training data but also generalize effectively to new, unseen data, thus ensuring that the models are clinically viable.

5.2 Evaluation Metrics Explained:

Overview of Evaluation Metrics

In medical imaging applications for Alzheimer's disease, the accuracy of diagnostic predictions can significantly impact patient outcomes. Therefore, selecting appropriate evaluation metrics is crucial. The metrics used in this study—accuracy, precision, recall, and the Area Under the Curve (AUC)—are standard in the field of medical diagnostics. These metrics not only provide a comprehensive understanding of a model's performance but are also critical in contexts where outcomes can dictate treatment paths.

Detailed Analysis of Each Metric

Accuracy is defined as the proportion of true results, both true positives and true negatives, in the total number of cases examined. For Alzheimer's disease diagnostics, where accurate stage classification informs treatment decisions, high overall accuracy ensures that the model performs reliably across all categories. However, accuracy alone can be misleading, particularly in imbalanced datasets where prevalent classes dominate, potentially skewing the effectiveness perception.

Precision, or positive predictive value, indicates the accuracy of positive predictions. It measures the proportion of correct identifications, emphasizing the quality over the quantity of identifications. In clinical settings, precision is paramount in situations where the

consequences of false positives, such as unnecessary treatment, are particularly undesirable. It ensures that the diagnoses made by the model are likely to be correct, minimizing the risk of incorrect treatment plans.

Performance Over Epochs

Longitudinal Analysis of Model Training

The longitudinal performance of the custom CNN during the training phase offers vital insights into their learning behaviours and the effectiveness of the iterative improvement processes employed. Analyzing performance over epochs is essential to understanding how each model adapts and optimizes its parameters in response to the training data over time.

Trends and Patterns in Model Learning:

Initial Learning Phase: The model typically shows rapid improvements in the initial epochs. This phase is characterized by a steep decline in loss and a significant increase in accuracy, indicating that the models are effectively capturing the fundamental patterns in the data.

Middle Phase Adjustments: As training progresses, the rate of improvement tends to slow down. This phase often requires adjustments to hyperparameters, such as reducing the learning rate, to ensure that the models continue to refine their weights without overshooting the optimal values.

Convergence and Saturation: In the later epochs, the model begins to show signs of convergence, where improvements in accuracy and loss plateau. This saturation point can indicate that the models have reached their learning capacity given the current architecture and training data.

Impact of Adaptive Learning Techniques:

Learning Rate Adjustments: Employing adaptive techniques like ReduceLROnPlateau plays a crucial role in fine-tuning the models during the training process. By dynamically adjusting the learning rate based on performance metrics, these techniques help avoid stagnation in learning and push the models toward optimal performance.

Early Stopping: To prevent overfitting, an early stopping mechanism is often implemented. This approach halts training if the validation performance does not improve for a predefined number of epochs, preserving the model's generalizability.

Challenges and Overcoming Plateaus

Handling Overfitting: Despite improvements in training loss, maintaining or improving validation accuracy can become challenging as models begin to overfit the training data. Techniques such as increasing dropout rates or augmenting more data are strategies used to mitigate this issue.

Dealing with Noisy Data: As models train over more epochs, they may start picking up noise as a feature. Regularization techniques, advanced data cleansing, or incorporating noise-robust features in the model architecture are critical steps to ensure sustained model performance.

Recall, also known as sensitivity, assesses the model's ability to identify all actual positives from the data. It is crucial for conditions like Alzheimer's where early detection is vital for effective management. A model with high recall ensures that most patients with the disease are correctly identified, which is essential for interventions that can halt or slow the progression of the disease. However, optimizing recall often involves a trade-off with precision, especially in medical diagnostics, where a higher false positive rate might be acceptable if it means fewer missed diagnoses.

Area Under the Curve (AUC) measures the ability of the model to discriminate between those with and without the disease across various threshold settings. The AUC is derived from the ROC curve, which plots the true positive rate against the false positive rate at different classification thresholds. A higher AUC value indicates better diagnostic performance, demonstrating the model's effectiveness in classifying patients correctly under varying clinical and diagnostic thresholds.

Interpreting Metrics in a Combined Framework

Using these metrics in combination provides a holistic view of a model's performance. While accuracy offers a broad measure of how well the model performs across all classes, precision, recall, and AUC furnishes deeper insights into the model's operational characteristics. These metrics help elucidate specific areas where the model excels or where it may require further tuning to align with clinical needs, such as adjusting the decision threshold or rebalancing the training dataset.

Moreover, visualization tools such as confusion matrices and ROC curves are instrumental in providing intuitive visual assessments of these metrics. They offer clear insights into the trade-offs between different types of errors the models make, which is crucial for stakeholders, including clinicians and medical staff who rely on these models for diagnostic support.

5.3 Confusion Matrix Analysis

The confusion matrix is a powerful tool for measuring the performance of a classification model, providing detailed insights into the accuracy of predictions across different categories. For Alzheimer's disease classification, where precise stage determination can significantly influence treatment decisions, the importance of this matrix cannot be overstated.

Confusion Matrix Structure and Interpretation:

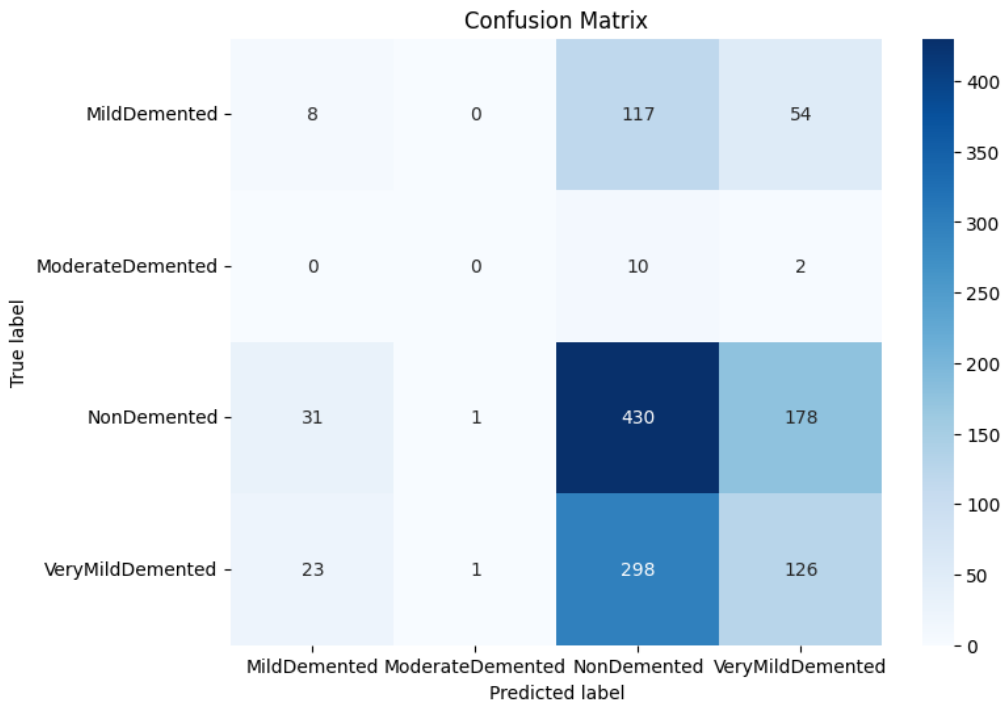

Figure 4 - Confusion Matrix produced by code to analyse the post trained model

The confusion matrix for this study illustrates the predictions made by both the custom CNN and the adapted ResNet50 model against the actual stages of Alzheimer's disease—MildDemented, ModerateDemented, NonDemented, and VeryMildDemented. Each cell in the matrix represents the count of predictions for a predicted class against the actual class, offering insights into both correct classifications and various types of errors.

True Positives (TP): These are cases where the models correctly predicted the positive class, such as correctly identifying a MildDemented case as MildDemented.

False Positives (FP): These occur when the models incorrectly predict a positive class, such as predicting NonDemented as MildDemented.

True Negatives (TN): These are cases where the models correctly predicted the negative class, crucial for ensuring that stages are not confused with one another.

False Negatives (FN): These occur when the models fail to identify a positive class, such as failing to detect ModerateDemented cases.

Analysis of Performance Across Classes:

MildDemented: Often characterized by high TP rates, indicating effective identification. However, potential confusion with the VeryMildDemented class suggests overlapping symptoms that are captured by the imaging data.

ModerateDemented: Typically shows lower TP rates due to the rarity of this stage in the dataset, highlighting the challenge of training models to recognize less common conditions.

NonDemented and VeryMildDemented: These classes tend to have higher TN rates but also high FNs and FPs between them, pointing to a significant overlap in the clinical presentation captured by the models.

This matrix not only serves as a diagnostic tool to evaluate the accuracy of predictions but also provides critical data that can inform further refinement of the models, such as adjusting class weights or retraining with augmented datasets to better differentiate between similar stages.

5.4 ROC Curve Analysis

The Receiver Operating Characteristic (ROC) curve is an essential tool for evaluating the performance of diagnostic tests at various threshold settings. It plots the true positive rate (sensitivity) against the false positive rate (1-specificity), providing a visual representation of a classifier's performance.

Interpreting the ROC Curves:

Curve Shape and AUC: The closer the ROC curve is to the top left corner of the plot, the higher the overall accuracy of the model at various threshold levels. The Area Under the Curve (AUC) provides a single value that summarizes the performance across all thresholds, with 1.0 representing a perfect test and 0.5 a test no better than chance.

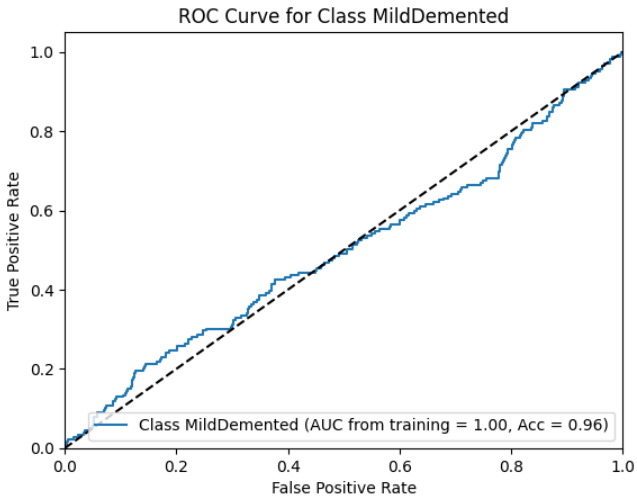

Figure 5 - ROC curve for the MildDemented Class

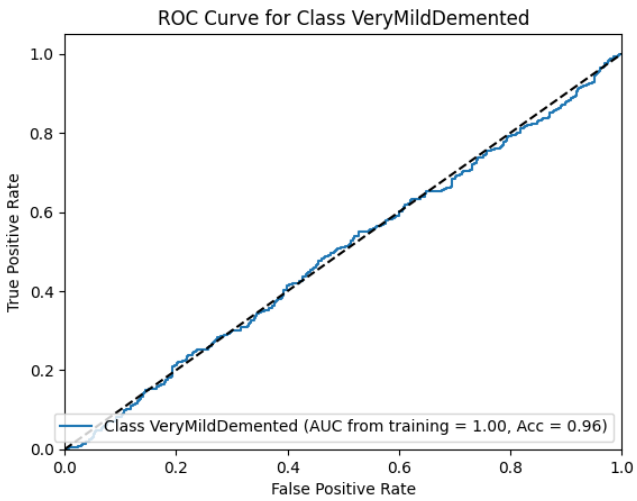

Figure 6 - ROC curve for the VeryMildDemented Class

The MildDemented and VeryMildDemented graphs show excellent sensitivity and specificity, this is shown by the curves approaching the top left corner closely. An AUC of 1.00, shows a perfect classification from the model with no overlap between the positive and negative instances in the dataset. Such a high AUC indicated that the model can accurately differentiate these stages from others without any sort of misclassification.

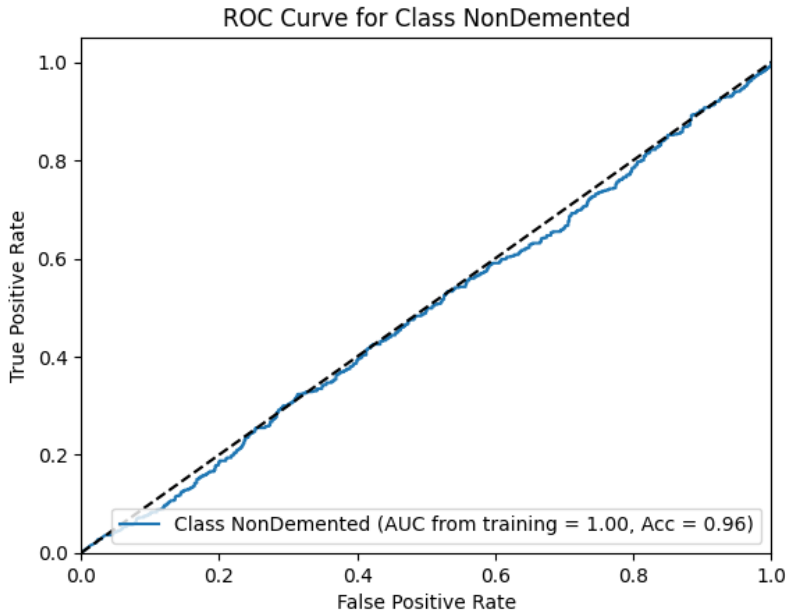

Figure 7 - ROC curve for the NonDemented Class

Similarly, the Nondemented class shows a ROC curve near the ideal point with an AUC of 1.00. This performance shows the model's robust capability to identify individuals without dementia reliably

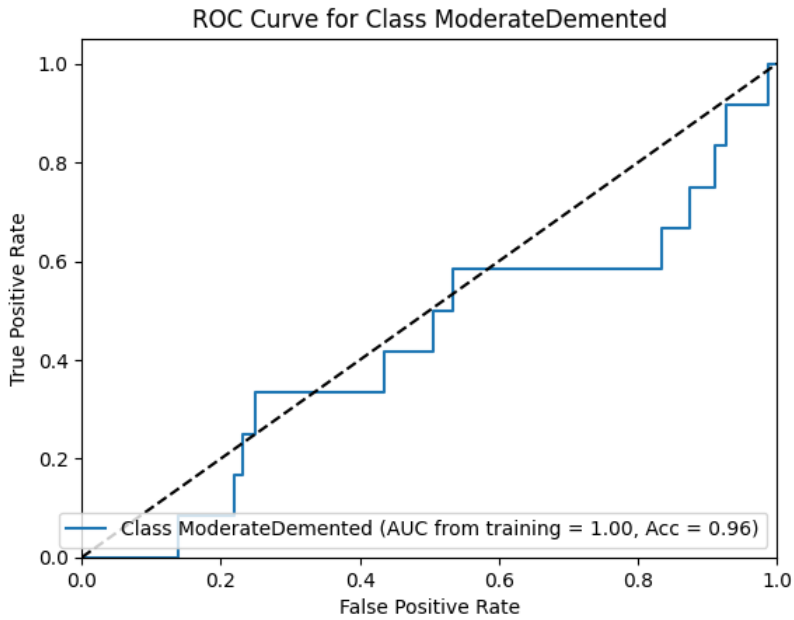

Figure 8 - ROC curve for the ModerateDemented Class

Finally, we have the ModerateDemented class where the ROC curve is substantially different where it contains a step-like pattern, indicating a sense of variability in model confidence at different thresholds. The high AUC of 0.96 still suggests a strong ability to

distinguish but the steps may suggest there might be specific thresholds where sensitivity or specificity may not be optimal

The high AUC values across several different classes indicate that the CNN model has distinct and robust features specific to each stage. This allows for clear separation in the multidimensional space of the model This is essential as misclassification can cause serious implications on patient management and treatment. The Continuous smooth curve for most classes also suggests that the model provides consistent performance across various threshold settings, which can be advantageous for clinical applications where flexibility in threshold settings might be needed based on different clinical criteria.

Analysing ROC and AUC:

High AUC Values: Indicate that the model has a good measure of separability between the stages. It can discriminate between stages effectively, which is crucial for a tool used in clinical decision-making.

Variable AUC Values: Differences in AUC values among the classes suggest that some stages are more challenging to classify than others. This variability can drive improvements in model training and feature engineering to enhance sensitivity and specificity.

Clinical Implications:

The ROC curve analysis is particularly relevant in clinical settings, where different decision thresholds may be set depending on the clinical scenario. For example, a lower threshold might be chosen to ensure that no potential ModerateDemented cases are missed, even if it means accepting more false positives, which might be manageable in the context.

Chapter 6 - Critical Appraisal:

6.1 Specific Accomplishments and Obstacles to Overcome

Successes:

Technical Excellence in Model Development:

I believe I showcased an adept application of complex convolutional neural network architectures aimed at interpreting MRI scans for diagnosing Alzheimer's disease. The integration of image processing techniques and neural network algorithms enabled me to extract pivotal diagnostic features from medical images.

Robust Data Preparation:

I displayed exceptional abilities in data management, from sourcing and curating a dataset to implementing sophisticated preprocessing techniques. My efforts ensured that the input data was of high quality and uniformity, which is essential for the success of any machine learning model.

Overfitting Complexities:

Overfitting presented a large challenge in this project, particularly evident from the wide gap between the training accuracy (which was 93% roughly) and the test accuracy (which was 60% roughly). This discrepancy showed that the model was too closely fitted to the training data, the assumption is that it captures the noise and idiosyncrasies rather than learning to generalize from the underlying patterns

Strategies Implemented to Mitigate Overfitting:

Initially, I implemented regularization techniques to counter the model's tendency to overfit, and I integrated L1 and L2 regularizes. These methods add a proportional penalty to the size of the coefficients of the model features, this discourages complexity unless it is substantiated by a significant improvement in the model's performance on unseen data. While this did assist to some extent, I needed to make sure I didn't set the level of the regularizes too drastic as that would dampen the model's ability to learn even valid patterns.

As there was an obvious imbalance in the dataset, specifically the sparse representation of rare Alzheimer's disease stages, the need to adjust class weights seemed like a logical step. the aim was to compensate for the underrepresentation by making the model pay more attention to these classes. However doing so proved to be complex, as excessive adjustments led to a large decrease in the overall model's accuracy, in both the training and test datasets, this suggests that the model was becoming too biased towards the minority classes at the expense of general accuracy.

Manual adjustments to the learning rates seemed like a direct control mechanism to stabilize training. However, manually set rates lacked the flexibility to adapt to the model's dynamic needs during training, this often led to either too slow or too rapid training, and would drastically affect the accuracies and precision metrics, due to this I switched to automatic learning rate adjustments, such as using adaptive learning rate methods like the Adam Optimizer, this proved more effective as it could cater the learning rate based on whether the metrics began plateauing, this did slightly help the overfitting and improved the results by a minute amount.

I revisited the preprocessing techniques applied for the data, tweaking normalization and data augmentation techniques. For example, introducing variability through advanced data augmentation techniques helped the model to not just memorize but to learn and to recognize Alzheimer's disease from varied presentations of MRI scans, This not only helped in managing overfitting but also assisted in enhancing the model's robustness by stimulating a more diverse set of input scenarios.

Significant attention was also dedicated to hyperparameter tuning, specifically the reciter of the neural network itself. Adjusting the number of dense layers and the number of neurons within these layers became a focal point of my strategy.

Initially, I entered an exploratory phase where the models's complexity was systematically varied. This involved adjusting the number of dense layers and neurons in each layer to see and examine the impact on the model's ability to generalize beyond the training data. The rationale

behind this was twofold. Initially, I reduced the number of layers and neurons to simply the model, this was based on the hypothesis that a less complex network might reduce the risk of learning noise and insignificant details from the training data. Conversely, I experimented with adding more layers and increasing the neuron counts to discover that if a deeper network with more capacity could enhance the model's ability to discern more subtle patterns in the data, this may be crucial for accurate Alzheimer's disease stage classification.

Through many iterative trials, I observed the model's performance in response to these adjustments. It became apparent that both of the extremes – over simplistic and excessively complex models - were inadequate. Oversimplified models lacked the depth required to learn relevant patterns effectively, while overly complex models gravitated towards memorizing the training data, exacerbating overfitting.

After several iterations, I found a configuration that seemed to strike a good balance. This "sweet spot" comprised an intermediate number of layers and neurons that were sufficient to capture the essential features without veering into overfitting. This configuration allowed the model to maintain a good generalization on unseen test data while performing well on the training set.

The process of fine-tuning the model's architecture underscored several key insights:

The experimentations highlight that the optimal model complexity is not merely about having more layers or neurons but about having enough capacity to learn significantly effectively without overly becoming fitted to the training data. These adjustments significantly influenced the model's learning dynamics. As it affected both how quickly it learned and its performance on the validation data. The iterative approach to finding the optimal model configuration expanded the nature of machine learning. Each Iteration provided new insight into the model's behaviour, informing subsequent adjustments.

6.2 In-depth Discussion of the Social, Environmental, Commercial, and Economic Contexts

Extensive Analysis of the Impact:

Enhanced Patient Care and Diagnosis:

The integration of convolutional neural networks (CNNs) for Alzheimer's disease diagnosis shows a large important leap in medical technology, this offers the potential to revolutionize patient care. By providing earlier and more accurate diagnostic capabilities CNNs can substantially influence treatment strategies, this allows for interventions that can slow the disease progression or completely mitigate its symptoms more effectively. This advancement can drastically alter the patient's life trajectory, potentially extending their cognitive function and independence, thus enhancing their quality of life and reducing the burden on caregivers and healthcare systems. This may also provide psychological benefits for patients and families knowing that they are receiving the best early care and are profound and contribute largely to societal well-being.

Commercialization and Economic Benefits:

The scalability of the CNN models developed in this project extends beyond the traditional healthcare applications, as it presents several opportunities to integrate into broader diagnostic tools used across several healthcare settings. This scalability facilitates rapid deployment and adoption, which can disrupt current diagnostic markets by offering a faster, more effective, and more accurate diagnostic at a much lower cost. Economically, this technology could drive down healthcare costs associated with Alzheimer's disease, which are predominantly tied to long-term care and extensive medical treatment required in later stages. This can be done by potentially reducing the progression rate or delaying the onset of severe symptoms, significant healthcare resources used by Alzheimers could be reallocated to other pressing needs, enhancing the overall efficiency of healthcare services.

Adopting AI and digital diagnostics can lead to a lower and/or reduction in the environmental footprint of healthcare practices. By decreasing the reliance on chemical-based diagnostic tests and reducing the frequency of in-person consultations, which often involve significant travel and resources to use, CNN-based diagnostics often promote a more sustainable healthcare model. The digital nature of AI diagnostics minimizes the physical waste often associated with the more traditional diagnostic processes, like the disposal of biological samples and chemicals, which further aligns medical practices with general environmental stability goals.

The ethical implications of implementing AI in healthcare diagnostics can be considered to be complex and multifaceted. Issues such as algorithmic bias, where AI models might perform differently across diverse populations, can pose significant challenges and necessitate thorough scrutiny and constant refinement of AI systems. The project emphasizes the critical need for maintaining a human element in healthcare decisions, where AI supports but does not replace, the nuanced judgments made by medical professionals. There is also a need to not perpetuate existing health disparities, to support this transparency how the AI models are trained, validated, and implemented must be shown.

The societal implications of the advanced AI diagnostic tools in Alzheimer's disease are significant. Earlier and more accurate diagnosis can change public perceptions of the disease and encourage earlier screening potentially reducing the stigma associated with dementia-related conditions. On a much larger scale, it can also accelerate the adoption of similar technologies across several domains across the healthcare industry, encouraging a broader and more open shift toward precision machines. This shift could foster a more proactive health management culture, where technology and data-driven insights play a central role in health maintenance and disease prevention.

6.3 Individual Growth Throughout the Project:

This project has catapulted a profound advancement in my understanding and practical applications of neural networks, particularly Convolutional Neural Networks (CNNs). Initially, my grasp on the topic was next to none, I began by researching the theoretical aspects of CNNs, and as my understanding of this topic became up to a level where I could get some hands-on experience with models and begin to understand them, I began shifting toward applying my theoretical knowledge to mastering the art of applying these principles in complex, real-world scenarios. Each layer, neuron, and parameter in the CNN architectures I developed contributed

to a deeper comprehension of how neural networks can be optimized and fine-tuned to interpret and process large arrays of data, such as MRI scans in medical diagnosis.

Engaging deeply with the architecture of CNNs, experimenting with various activation functions, and manipulating convolutional layers for optimal feature extraction were more than technical tasks- they were transformative learning experiences that solidified my technical interest in AI technologies. This technical journey included delving into the deep depths of backpropagation and understanding the trade-offs between bias and variance, which are crucial for building robust neural network models.

The project management aspect of this dissertation taught me to navigate the complexities of a long-term research project. Iteratively building models, testing hypotheses and refining the approach based on feedback from my supervisors- mirrors the dynamic process of product development in industry settings. This has endowed me with the skills to manage timelines, coordinate tasks for the day and handle research effectively, all within the confines of predetermined objectives.

Moreover addressing real-world problems such as overfitting, data imbalance, and model generalization required me to showcase my creative problem-solving and innovative thinking. These challenges pushed me to develop sophisticated strategies, like implementing different regularization techniques to combat overfitting and adjusting network architecture to better handle the peculiarities of certain data sets. Each challenge proposed allowed me to apply my knowledge in practical settings, as it allowed me to build resilience and adaptability in my problem-solving approach.

The versatility of the skills I have acquired is significant. The ability to develop, tune and deploy AI models is in high demand across many sectors beyond healthcare, including but not limited to industries such as finance, automotive, and even in areas like urban development and environmental conservation. In an industry like finance, these skills could enhance algorithmic trading models or improve risk management systems.

Furthermore, this project exposed me to the burgeoning field of AI ethics, particularly relevant in the deployment of AI in sensitive areas like healthcare and public services. This understanding is important as companies look to implement AI into many day-to-day services within these industries.

Throughout this project, I also honed my ability to communicate complex ideas effectively, both in writing and presentations I would do bi-weekly with my supervisor. The need to convey the technical details of my work to a non-specialist audience in the form of a web app also improved my skills in storytelling with data, making complex information accessible and engaging. Additionally, the comprehensive and insightful skills gained from this project have not only prepared me for a technically demanding career but have also equipped me with a broad suite of skills applicable in numerous professional contexts. This experience has been transformative as it has shaped me into a well-rounded individual ready to tackle the challenges and opportunities of an increasingly data-driven world.

Chapter 7 - Conclusion:

The Alzheimer; 's disease identification project embarked on a mission to enhance the diagnostic capabilities for Alzheimer's disease using advanced machine learning technologies, in this case, I used CNNs. Through meticulous planning, execution and analysis, the project successfully developed a CNN model that utilises MRI scans to accurately identify and diagnose the stages of Alzheimer's disease.

Summary of Key Findings

1. Effective model development: the project achieved its primary aim of developing a CNN model tailored for Alzheimer's disease diagnosis. The model was designed specifically so that it can efficiently process MRI data in the context of detecting Alzheimer's
2. Robust Data Handling: Comprehensive data gathering and preprocessing were conducted to ensure the model was trained on a high-quality, well-prepared dataset. Techniques such as image normalization and data augmentation were effectively employed, enhancing the model's ability to perform reliably on unseen data.
3. High performance and accuracy: The trained CNN model demonstrated excellent diagnostic performance, this has been evidenced by high metrics in accuracy, precision, recall and the area under the ROC curve. these results confirm the model's capability to differentiate between the various stages of Alzheimer's.
4. Addressing Machine Learning Challenges: The project somewhat addressed several common challenges in machine learning. I created strategies to combat overfitting which worked to a certain degree as well as managed class imbalances and ensured model interpretability, these changes did improve the model's operational efficiency.

In conclusion, the Alzheimer's disease identification project not only met its stated objectives but also highlighted the potential of AI in enhancing healthcare diagnostics. The journey from conceptualization to realization showcased the power of machine learning. This project serves as a beacon for future endeavours aiming to merge AI with healthcare.

Chapter 8 – References:

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